

Topic 5 Summary: Force and Newton's Laws

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Force & Newton's Laws.

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→ For a body to change its motion, a force must act on the body.

Newton's First Law!

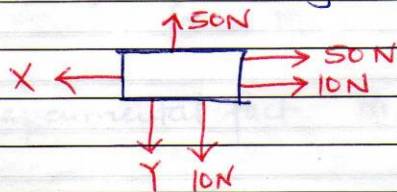
Every body continues in its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by forces impressed upon it.

A body at rest

If forces act on a body and it doesn't move, the forces must balance. Hence if a no. of forces act on a body and it remains at rest, the resultant force in any direction must be zero.

Example 1

A body remains at rest when subjected to the forces shown in the diagram. Find X and Y .



Solution

Horizontal forces balance $\Rightarrow X = 50 + 10 = 60\text{N}$

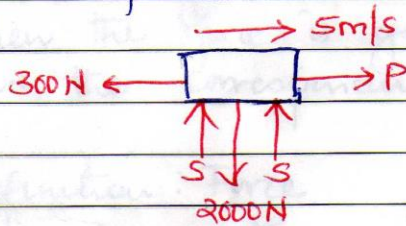
vertical " $\Rightarrow Y + 10 = 50 \Rightarrow Y = 40\text{N}$

A body in motion

A body can only change its velocity i.e. increase or decrease speed or change direction, if a resultant force acts upon it. Thus if it moves with constant velocity, there is no resultant force acting on it.

Example 2:

A body moves horizontally at a constant 5 m/s subject to the forces shown. Find P and S.



There is no vertical motion $\therefore 2S = 2000 \text{ N}$
 $S = 1000 \text{ N}$

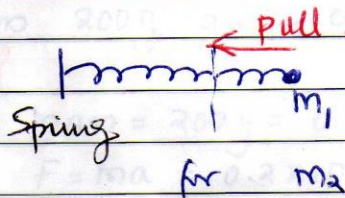
Horizontal velocity is constant $\therefore$ Resultant force is zero

i.e. $P = 300 \text{ N}$.

Mathematically, Newton's first law is written as

$$\frac{dv}{dt} = 0$$

Newton's Second Law.



$$\text{acceleration} = a_1 = a_2$$

Experimental fact $m_1 a_1 = m_2 a_2$.

Second law states:

The change of motion is proportional to the force impressed and is made in the direction of the straight line in which that force is impressed.

OR
 The rate of change of momentum is directly proportional to the applied force and takes place in the same direction as the force.

$$\text{i.e. } F \propto \frac{d}{dt}(mv)$$

$$F = km \frac{dv}{dt}$$

but $k=1$

$$F = k \frac{d}{dt}(mv)$$

$$\therefore \boxed{\vec{F} = m\vec{a}}$$

Newton's 2nd law gives us an explicit expression for force and gives us a mechanism to evaluate it. Also when the force is given, it provides a way to find the corresponding acceleration.

Definition: Force

Force is a vector quantity. The magnitude total force $\vec{F}$ acting on an object is the product of its mass with its acceleration. The direction of $\vec{F}$ is the direction of the acceleration. Thus

$$\vec{F} = m\vec{a}$$

units: $\text{kgm/s}^2 = \text{Newton}$ i.e. $1\text{N} = 1\text{kgm/s}^2$

Example 3:

Find the resultant force that would give a body of mass 200g an acceleration of 10m/s^2 .

Sol

$$m = 200\text{g} = 0.2\text{kg}$$

$$F = ma = 0.2 \times 10 = \underline{2\text{N}}$$

Example 4:

Find in vector form, the acceleration produced in a body of mass 5kg subject to forces $(4\hat{i} + \hat{j})\text{N}$ and $(-\hat{i} + \hat{j})\text{N}$. Also state the magnitude and direction of the acceleration.

Sol

$$\begin{aligned}\text{Resultant } \vec{F} &= (4\hat{i} + \hat{j}) + (-\hat{i} + \hat{j}) \\ &= 3\hat{i} + 2\hat{j} \text{ N.}\end{aligned}$$

$$\begin{aligned}\vec{F} &= m\vec{a} \Rightarrow 3\hat{i} + 2\hat{j} = 5\vec{a} \\ \vec{a} &= \frac{3}{5}\hat{i} + \frac{2}{5}\hat{j}\end{aligned}$$

$$|\vec{a}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2} = 0.72 \text{ m/s}^2 \quad \text{direction} = \tan^{-1}\left(\frac{2/5}{3/5}\right) = \underline{33.7^\circ}$$

Example 5:

A particle of mass 5 units moves along a space curve defined by $\vec{r} = (4t^3 - t)\hat{i} + (6t^2 + 5t)\hat{j} + (2t^3 - t + 4)\hat{k}$. Find the velocity, acceleration and force acting on the particle when

(i) $t=0$

(ii) $t=2$

Sol.

$$\vec{r} = (4t^3 - t)\hat{i} + (6t^2 + 5t)\hat{j} + (2t^3 - t + 4)\hat{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = (12t^2 - 1)\hat{i} + (12t + 5)\hat{j} + (6t^2 - 1)\hat{k}$$

$$\text{at } t=0, \vec{v}_0 = (12(0) - 1)\hat{i} + (12(0) + 5)\hat{j} + (6(0) - 1)\hat{k} \\ = \underline{-\hat{i} + 5\hat{j} - \hat{k}}$$

$$\text{at } t=2, \vec{v}_2 = (12(2^2) - 1)\hat{i} + (12(2) + 5)\hat{j} + (6(2^2) - 1)\hat{k} \\ = \underline{47\hat{i} + 29\hat{j} + 23\hat{k}}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 24t\hat{i} + 12\hat{j} + 12t\hat{k}$$

$$\text{at } t=0, \vec{a} = 12\hat{j}$$

$$\therefore \vec{F} = m\vec{a} = 5 \times 12\hat{j} = \underline{60\hat{j}}$$

$$\text{at } t=2, \vec{a} = 48\hat{i} + 12\hat{j} + 24\hat{k} \\ \vec{F} = \underline{240\hat{i} + 60\hat{j} + 120\hat{k}}$$

Example 6:-

A body of mass 5 kg is moving with a velocity $\begin{pmatrix} 46 \\ 0 \end{pmatrix} \text{ m/s}$ when it is acted upon by a force $\begin{pmatrix} -30 \\ 35 \end{pmatrix} \text{ N}$. Find the magnitude and direction of its velocity 3s after the force began to act.

Sol:

$$\vec{F} = m\vec{a}$$

$$\begin{pmatrix} -30 \\ 35 \end{pmatrix} = 5\vec{a} \Rightarrow \vec{a} = \begin{pmatrix} -6 \\ 7 \end{pmatrix}$$

$$\vec{v} = \vec{u} + \vec{a}t$$

$$= \begin{pmatrix} 46 \\ 0 \end{pmatrix} + \begin{pmatrix} -6 \\ 7 \end{pmatrix} 3$$

$$= \begin{pmatrix} 28 \\ 21 \end{pmatrix} \text{ m/s}$$

$$|\vec{v}| = \sqrt{28^2 + 21^2} = \underline{\underline{35 \text{ m/s}}}$$

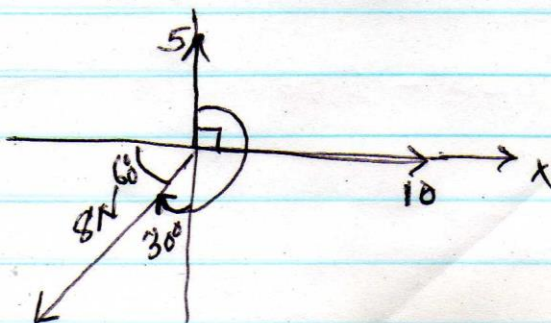
$$\tan \theta = \frac{21}{28} \quad \theta = \underline{\underline{36.87^\circ}}$$

velocity is 35 m/s in the dir. 36.87° from the +ve x-axis.

Example 7:

Given forces 5N, 10N, 8N in the direction with bearings $000^\circ, 090^\circ, 210^\circ$ respec. find the magnitude & bearing of the resultant.

Sol.



In terms of $\hat{i}$ and $\hat{j}$ i.e. in x and y directions, the forces can be expressed as :

$$5\text{N} : \begin{pmatrix} 0 \\ 5 \end{pmatrix} = 5\hat{j}$$

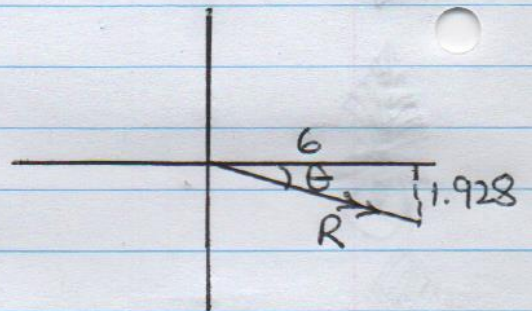
$$10\text{N} : \begin{pmatrix} 10 \\ 0 \end{pmatrix} = 10\hat{i}$$

$$8\text{N} : \begin{pmatrix} -8\cos 60^\circ \\ -8\sin 60^\circ \end{pmatrix} = -4\hat{i} - 4\sqrt{3}\hat{j}.$$

$$\begin{aligned}\vec{R} &= 5\hat{j} + 10\hat{i} + (-4\hat{i} - 4\sqrt{3}\hat{j}) \\ &= 6\hat{i} - 1.928\hat{j}\end{aligned}$$

$$|\vec{R}| = \sqrt{6^2 + (1.928)^2} = \underline{6.3\text{N}}$$

$$\theta = \tan^{-1} \left(\frac{-1.928}{6} \right) = \underline{-17.83^\circ}$$



The resultant force is $\vec{F} = 6.3\text{N}$ in the direction -17.83° from the positive x-axis.

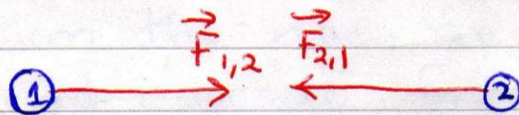
Newton's Third Law:

Newton realised that when two bodies interact via a force, then the force on one body is equal in magnitude and opposite in direction to the force acting on the other body.

Law III: To every action, there is always an equal and opposite reaction.

In other words: The force which object A exerts on object B is equal in magnitude and opposite in direction to the force which B exerts on A.

Consider two bodies ^{1 and 2} engaged in a mutual interaction, as shown.



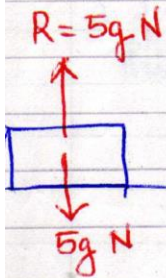
$F_{1,2}$ is the force on body 1 due to interaction with body 2, and $F_{2,1}$ is force on body 2 due to interaction with body 1.

Then these two vector forces are equal in magnitude and opposite in direction.

i.e. $\vec{F}_{1,2} = -\vec{F}_{2,1}$

Example 8: Normal Reaction

A box of mass 5 kg rests on a horizontal floor. The box exerts a force on the floor and the floor "reacts" by exerting an equal and opposite force on the box.



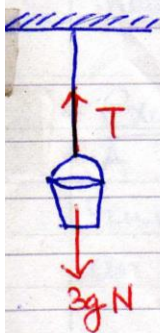
As the box is at rest, this reaction force R (or N) must equal the weight of the box.
i.e. $R = 5g \text{ N}$

This force R (or N) is called the **Normal Reaction** as it acts at right angles to the surfaces in contact.

Example 9: Tension

A bucket of mass 3 kg hangs on a vertical rope attached to a beam. The bucket exerts a force on the rope, so the rope exerts an equal and opposite force on the bucket. As the bucket is at rest, this force T must equal the weight of the bucket.

$$\text{i.e. } T = 3g \text{ N.}$$

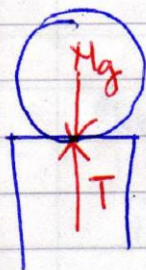


The force T is called the Tension in the rope.

E.g 10: Thrust

A granite sphere ^{mass} M kg rests on top of a pillar. The sphere exerts force on the pillar and pillar exerts equal & opp. force T on sphere.

$$\underline{T = Mg \text{ N}}$$



This force T is called the thrust in the pillar.

Gravity, Weight & Mass.

A body falling under gravity experiences an acceleration of 9.81 m/s^2 . From Newton's second law it is clear that this acceleration must be due to a force acting on the body.

This force is called the weight (W) of the body

$$a = g$$

$$F = W \quad \therefore F = ma \text{ gives } \boxed{W = mg}$$

A body of mass m has a weight mg N.

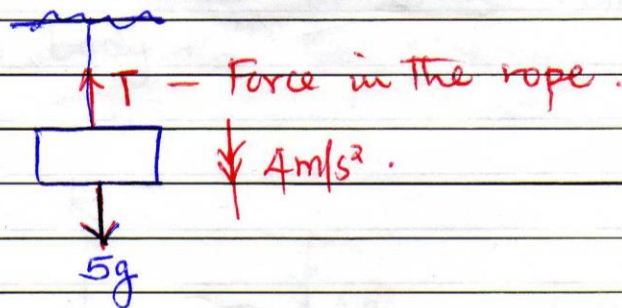
The mass of a body is a measure of its reluctance to change its motion when a force acts and is sometimes called its inertia.

Example 18:

A box of mass 5 kg is lowered vertically by a rope. Find the force in the rope when the box is lowered with an acceleration of 4 m/s^2 .

Soln

FBD:



The resultant vertical force on the box is $(5g - T) \text{ N}$ downwards

$$\begin{aligned} \text{i.e. } \vec{F} &= ma = 5g - T \\ &= 5 \times 4 = 5g - T \\ \therefore T &= 5g - 20 \\ &= \underline{\underline{29.05 \text{ N}}} \end{aligned}$$

$$(g = 9.81 \text{ m/s}^2)$$

Example 12:

A force of $5w$ acts vertically upwards on a body of weight $3w$. Find the acceleration in terms of g .

Sol:

$$\text{Resultant force} = 5w - 3w = 2w$$

$$\text{Weight} = Mg$$

$$\text{Mass of body of weight } 3w = 3w/g.$$

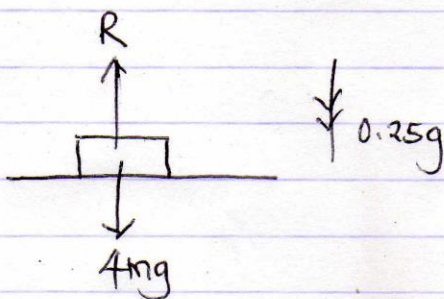
$$F = ma \Rightarrow a = F/m$$

$$= \frac{2w}{3w/g} = \underline{\underline{\frac{2g}{3} \text{ m/s}^2}}$$

Example 13:

A body of mass $4m$ stands on a platform which is accelerating downwards at $0.25g$. Find the reaction between the body and the platform.

Sol:



The resultant downward force $\neq$ from $F = ma$

$$F = 4m \times 0.25g = Mg$$

but Resultant downward force = Weight - Reaction

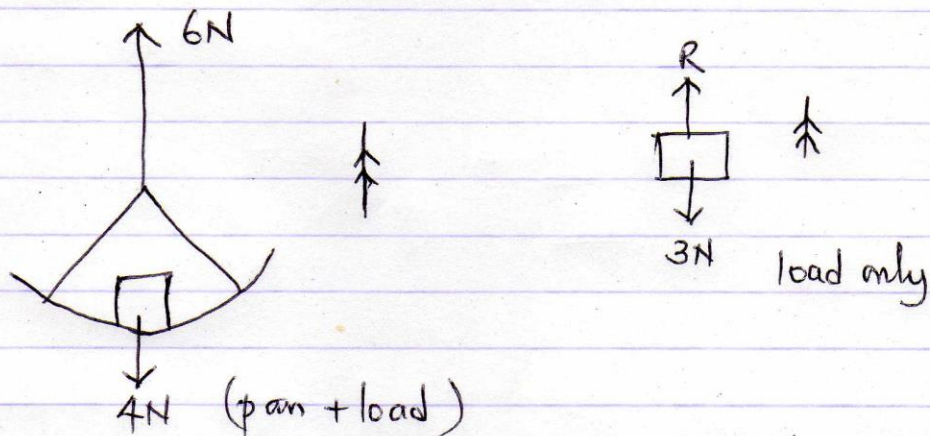
$$\text{i.e. } Mg = 4Mg - R$$

$$\therefore \underline{\underline{R = 3Mg}}$$

Example 14:

A pan of mass 100g carrying a body of mass 300g is lifted vertically upwards by a force of 6N. Find the reaction between the pan and the weight. ($g = 10 \text{ m/s}^2$).

Sol:



For pan + load

$$\text{Resultant upward force} = 6\text{N} - 4\text{N} = 2\text{N}$$

$$\text{Total mass} = 0.1 + 0.3 = 0.4 \text{ kg}$$

$$\vec{a} = F/m = 2/0.4 = \underline{\underline{5 \text{ m/s}^2}}$$

For ~~pan~~ load alone,

$$R - 3 = ma$$

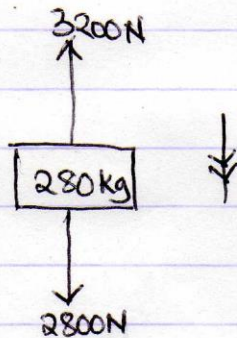
$$R - 3 = 0.3 \times 5 \Rightarrow R = \underline{\underline{4.5 \text{ N}}}$$

Example 15:

An 80 kg man is descending in a lift of mass 200 kg. While the lift is slowing down, the tension in the lift cable is 3200 N. Find

- the retardation of the lift.
- the force between the man and the floor of the lift.

Sol:



$$\begin{aligned} \text{a)} \quad F &= 3200 - 2800 = 400 \text{ N} \\ F &= ma \quad \therefore -400 = 280a \\ a &= \underline{\underline{-1.4286 \text{ m/s}^2}} \end{aligned}$$

$\therefore$ for the man,

$$\begin{aligned} 800 - R &= ma \\ 800 - R &= 80 \times -1.4286 \\ R &= 800 + 80(1.4286) \\ &= \underline{\underline{914.286 \text{ N}}} \end{aligned}$$

Force And Motion of Connected Bodies

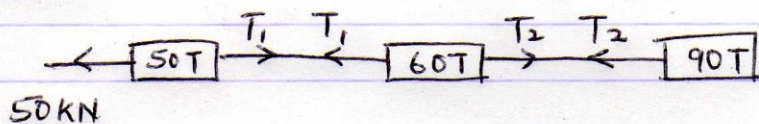
In the following examples, the strings are all considered to be massless and inextensible.

Also, the surfaces and pulleys are smooth, meaning there is no friction, or resistance to motion.

Example 1:

Three coupled trucks, with masses 50, 60, and 90 tonnes, are pulled along a horizontal track by a locomotive which applies a force of 50 kN to the first truck. Find the tensions in the couplings.

Sol:



The acceleration is equal throughout.

Considering, the whole system,

$$M = 50T + 60T + 90T = 200T$$

$$F = ma$$

$$50,000 = 200,000 a$$

$$\therefore a = \underline{0.25 \text{ m/s}^2}$$

To determine T_1 and T_2 , we take for T_1 , $F = ma$, where $m = 60 + 90 = 150T$. The two T_2 forces become internal forces which cancel each other out.

$$\therefore T_1 = 150,000 \times 0.25$$

$$= 37,500 \text{ N}$$

$$= \underline{37.5 \text{ kN}}$$

For T_2 , we apply $F=ma$ to the 90T truck alone (T_1 forces cancel each other out).

$$\therefore T_2 = 90,000 \times 0.25 \\ = 22,500 \text{ N} = \underline{22.5 \text{ kN}}.$$

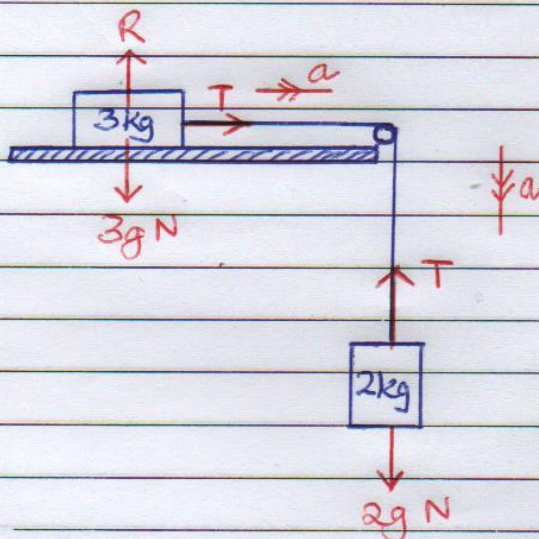
Example 2:

Consider a body of mass 3kg at rest on a smooth horizontal table. This body is connected by a light string which passes over a smooth pulley, as shown, to another body of mass 2kg hanging freely. What is the acceleration of the masses and the tension in the string?

Sol:

Since the string is massless & pulley is smooth, the tension in the string on both sides of the pulley will be the same.

The two bodies will have equal accelerations along the line of the string.



3kg mass

3 kg mass does not move in vertical direction $\therefore$

$$R = 3g$$

Horizontal force on 3 kg mass is T . Using $F=ma$,

$$T = 3 \times a \quad \text{--- (i)}$$

2 kg mass

$$2g - T = ma$$

$$2g - T = 2a \quad \text{--- (ii)}$$

Solving (i) & (ii)

$$2g - 3a = 2a$$

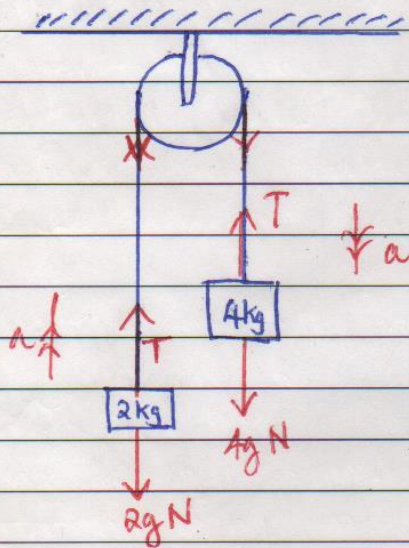
$$a = \underline{\underline{2/5g \text{ m/s}^2}}$$

$$\therefore T = 3 \times 2/5g = \underline{\underline{6/5g \text{ N}}}$$

Example 3:

Particles of mass 4 kg and 2 kg are connected by a massless string passing over a smooth fixed pulley as shown. The particles hang freely and are released at rest.

Find the acceleration of the two particles and the tension in the string.



For 2 kg mass

$$T - 2g = 2 \times a \quad \text{--- (i)}$$

For 4 kg mass

$$4g - T = 4a \quad \text{--- (ii)}$$

Eliminating T in (i) & (ii) gives

$$4g - 2g = 6a$$

$$2g = 6a$$

$$a = \frac{1}{3}g \text{ m/s}^2$$

$$\text{and } T = \frac{2}{3}g + 2g = \underline{\underline{\frac{8}{3}g \text{ N}}}$$

$$\text{Using } g = 9.81 \text{ m/s}^2 \text{ gives } a = 3.27 \text{ m/s}^2$$

$$T = 26.16 \text{ N}$$